notebook1

The purpose of this notebook is to document the development of the analysis. The final analysis will be run through a series of scripts, but I will develop those scripts interactively here and document my decision making process as I go.

```
# since this notebook is not in the main directory, use the here package
# to get proper file paths
library(here)
```

```
here() starts at /mnt/c/Users/jjk06/OneDrive/Desktop/2026_r_projs/hybrid_fitness_predict
```

```
library(ggplot2)
library(tidyr)
library(dplyr)
```

```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

Creating a file of seed sizes

Seeds were planted 20 at a time into grids of 4 rows by 5 columns. Before planting the seeds, they were placed onto a grid as they were to be planted and a photo taken with a scale in the photo. The areas of the seeds were then measured from the photograph. Four researchers were working on the project, to get used to using imageJ and make sure there was no bias, the seeds were measured by all four researchers.

First, I'm reading in the four files and just calculate the mean seed area for each seed.

```
initials <- c("jk", "js", "lb", "rp")

# there were some notes etc. beyond the 4th column, so only take the first 4 columns
tmp <- read.csv(here(paste0("data/raw/seed_areas_", initials[1], ".csv")))[,1:4]

tmp$researcher <- initials[1]

d <- tmp

for(i in 2:length(initials)){
  tmp <- read.csv(here(paste0("data/raw/seed_areas_", initials[i], ".csv")))[,1:4]

  tmp$researcher <- initials[i]

  d <- rbind(d, tmp)
}



# calculate the mean of the four measurements
seed_size <- d %>%
  group_by(tray, block, ind) %>%
  summarise(seed_area = round(mean(area),3))
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by tray, block, and ind.
i Output is grouped by tray and block.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(tray, block, ind))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

Next, I am going to attach the cross id and parentage to each individual.

```r
ids <- read.csv(here("data/raw/planting_ids.csv"))

# tray format is different in the two files, so need to change that to merge
ids <- ids %>%
  mutate(tray = gsub("LC", "lc_", tray)) %>%
  select(tray, block, cross, mother, father)

seed_size <- seed_size %>%
  left_join(ids) %>%
  mutate(cross = tolower(cross))
```

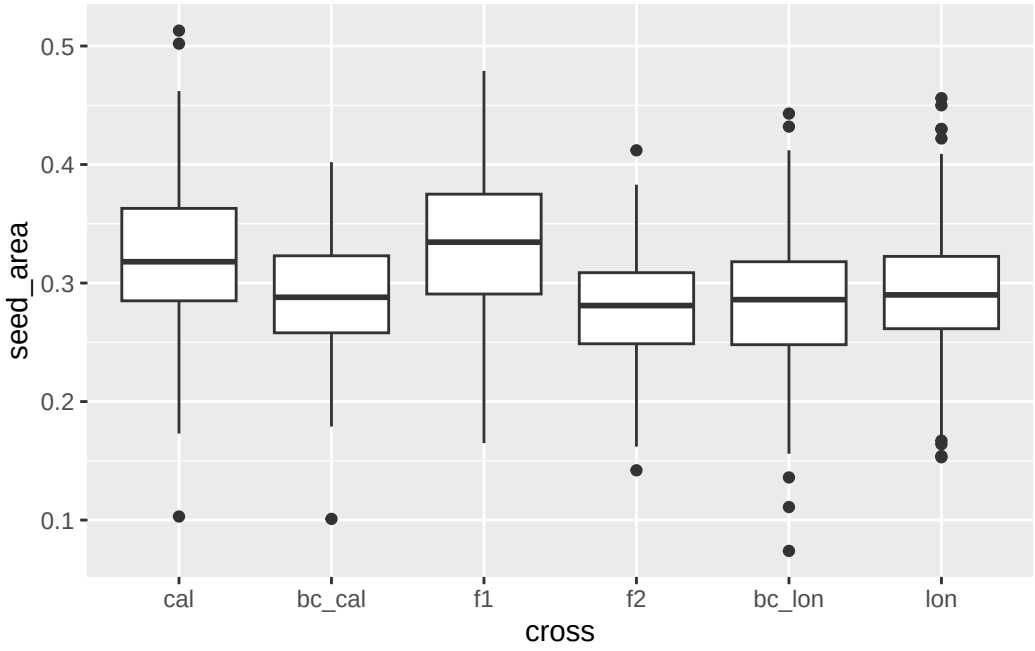
```
Joining with `by = join_by(tray, block)`
```

The above code takes the generates an object with seed sizes and cross information. It will be used in the script `scripts/get_seed_size.R` to generate the intermediate data file `reulsts/data/seed_size.csv`

Exploratory data analysis with seed size

First let's look at seed size vs cross type.

```r
seed_size %>%
  mutate(cross = tolower(cross),
         cross = factor(cross, levels = c("cal", "bc_cal", "f1", "f2", "bc_lon", "lon"))) %>%
  drop_na(seed_area) %>%
  ggplot(aes(x = cross, y = seed_area)) +
  geom_boxplot()
```



It looks like f1 and cal are the biggest with everything else being fairly similar. Let's just do a quick anova to see what that looks like.

```
summary(mod <- aov(seed_area ~ cross, data = seed_size))
```

```
              Df Sum Sq Mean Sq F value Pr(>F)
cross          5  0.594 0.11876   41.11 <2e-16 ***
Residuals   1370  3.957 0.00289
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
64 observations deleted due to missingness
```

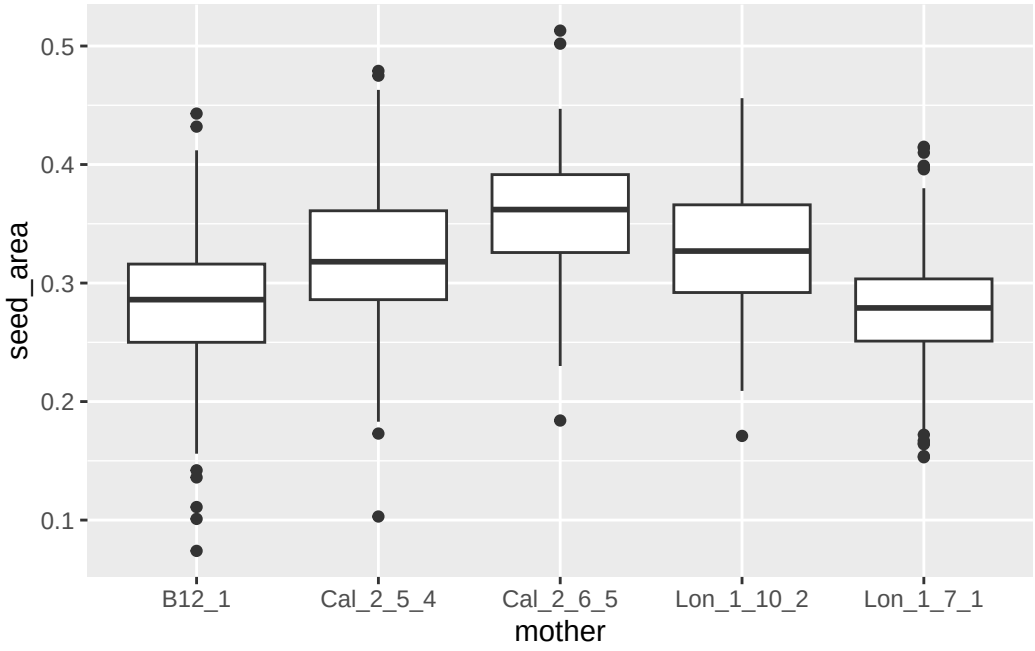
```
TukeyHSD(mod)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = seed_area ~ cross, data = seed_size)

$cross
                       diff           lwr          upr     p adj
bc_lon-bc_cal -0.004700301 -0.0190972135  0.009696612 0.9383445
```

```
cal-bc_cal     0.036615225  0.0223096875  0.050920762 0.0000000
f1-bc_cal      0.044222433  0.0298717261  0.058573140 0.0000000
f2-bc_cal     -0.009266329 -0.0237426551  0.005209996 0.4487747
lon-bc_cal     0.004501934 -0.0098640582  0.018867925 0.9479315
cal-bc_lon     0.041315525  0.0270739589  0.055557092 0.0000000
f1-bc_lon      0.048922734  0.0346357954  0.063209672 0.0000000
f2-bc_lon     -0.004566029 -0.0189791416  0.009847085 0.9455309
lon-bc_lon     0.009202234 -0.0051000571  0.023504526 0.4427199
f1-cal         0.007607208 -0.0065876466  0.021802063 0.6452916
f2-cal        -0.045881554 -0.0602033949 -0.031559713 0.0000000
lon-cal       -0.032113291 -0.0463235986 -0.017902983 0.0000000
f2-f1         -0.053488762 -0.0678557217 -0.039121803 0.0000000
lon-f1        -0.039720499 -0.0539762784 -0.025464720 0.0000000
lon-f2         0.013768263 -0.0006139641  0.028150490 0.0697293
```

So it seems like there might be some real differences here. I'm not so sure that it is because of the hybrid embryo though. Seed size is at least partially, if not mostly a maternal trait, so lets look at seed size vs seed parent.

```
seed_size %>%
  mutate(cross = tolower(cross),
         cross = factor(cross, levels = c("cal", "bc_cal", "f1", "f2", "bc_lon", "lon"))) %>%
  drop_na(seed_area) %>%
  ggplot(aes(x = mother, y = seed_area)) +
  geom_boxplot()
```



Looks like there are some differences, let's do a quick anova and see.

```
summary(mod <- aov(seed_area ~ mother, data = seed_size))
```

```
              Df Sum Sq Mean Sq F value Pr(>F)
mother         4  0.814 0.20339   74.61 <2e-16 ***
Residuals   1371  3.737 0.00273
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
64 observations deleted due to missingness
```

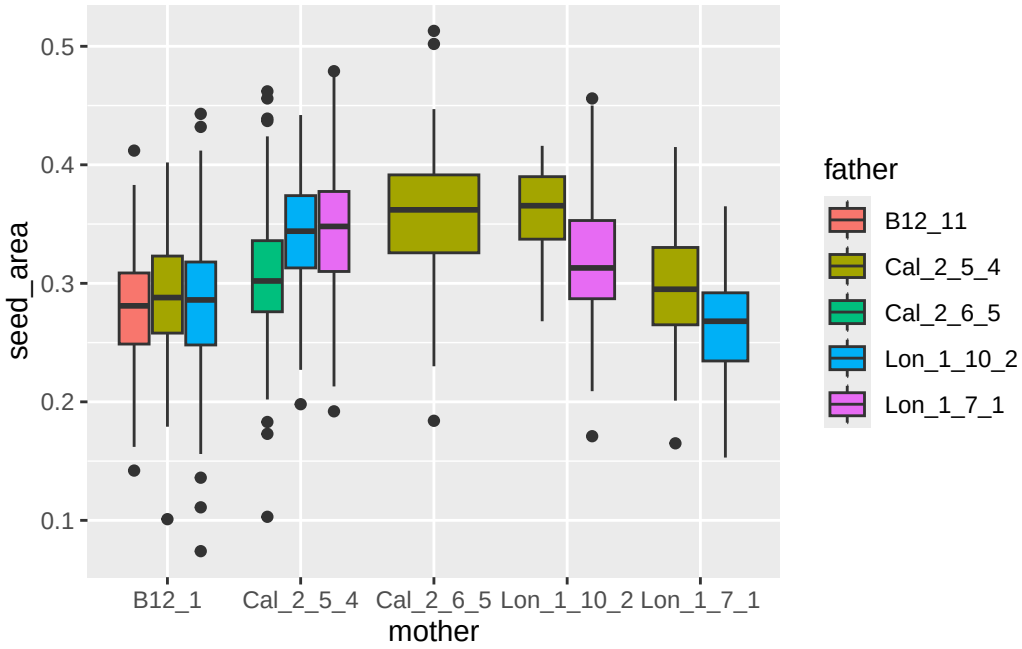
```
TukeyHSD(mod)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = seed_area ~ mother, data = seed_size)

$mother
                            diff          lwr          upr     p adj
Cal_2_5_4-B12_1       0.039812279  0.029589976  0.050034582 0.0000000
Cal_2_6_5-B12_1       0.073819605  0.056768466  0.090870745 0.0000000
```

```
Lon_1_10_2-B12_1       0.046663473  0.034032604  0.059294341 0.0000000
Lon_1_7_1-B12_1       -0.004230057 -0.015936444  0.007476330 0.8612564
Cal_2_6_5-Cal_2_5_4    0.034007326  0.015697695  0.052316957 0.0000044
Lon_1_10_2-Cal_2_5_4   0.006851193 -0.007433066  0.021135453 0.6849996
Lon_1_7_1-Cal_2_5_4   -0.044042337 -0.057516042 -0.030568632 0.0000000
Lon_1_10_2-Cal_2_6_5  -0.027156133 -0.046911789 -0.007400476 0.0016930
Lon_1_7_1-Cal_2_6_5   -0.078049663 -0.097227426 -0.058871899 0.0000000
Lon_1_7_1-Lon_1_10_2  -0.050893530 -0.066274820 -0.035512240 0.0000000
```

It looks like there might be some real differences between the mothers. Most of the parentals were crossed with an opposite parent as well, so let's add in paternal information to see how the embryo genotype affects seed size.

```
seed_size %>%
  mutate(cross = tolower(cross),
         cross = factor(cross, levels = c("cal", "bc_cal", "f1", "f2", "bc_lon", "lon"))) %>%
  drop_na(seed_area) %>%
  ggplot(aes(x = mother, y = seed_area, fill = father)) +
  geom_boxplot()
```



It seems that making crosses between parents increases seed size within the same mother, i.e., f1 embryos produce bigger seeds. The F1 mother produces the same size seeds whether crossed with another f1, lon, or, cal.

One last thing to do with seed size before moving on is that there is some missing data here. What happened is we tried to pre-apportion 20 seeds for each block ahead of time, but when we poured out the seeds or were arranging them to be photographed, we would sometimes loose a couple as they would jump around from static electricity or a misplaced sigh and then couldn't find them again. When that happened, we would leave the last places blank. So I just want to go through and figure out how many were missing from each block.

```
seed_size %>%
  drop_na(seed_area) %>%
  group_by(tray, block) %>%
  summarise(n = n())
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by tray and block.
i Output is grouped by tray.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(tray, block))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 72 x 3
# Groups:   tray [6]
   tray  block     n
   <chr> <int> <int>
 1 lc_1      1    15
 2 lc_1      2    20
 3 lc_1      3    17
 4 lc_1      4    20
 5 lc_1      5    20
 6 lc_1      6    20
 7 lc_1      7    20
 8 lc_1      8    20
 9 lc_1      9    20
10 lc_1     10    20
# i 62 more rows
```